Peripheral Troubleshooting Theory

1. List 5 common issues users face with USB keyboards or mice on laptops, and write one possible solution for each.

Here are 5 common issues users face with a USB keyboard or mouse on a laptop and their solutions:

1. **Device Not Recognized / Unresponsive (No lights or cursor movement)**
   - **Solution:** Unplug the USB cable or receiver, wait about 10 seconds for Windows to unload the driver, and plug it back into a different USB port.
2. **Erratic Cursor Movement or Laggy Typing**
   - **Solution:** Clean dust, debris, or blockages from the mouse sensor or underneath the keyboard keys using compressed air.
3. **Faulty or Outdated Device Drivers**
   - **Solution:** Open **Device Manager**, right-click the USB keyboard or mouse, and select **Update driver** or **Uninstall device**, then restart your laptop to let Windows automatically reinstall the proper driver.
4. **Power-Related Issues Using a USB Hub**
   - **Solution:** Plug the keyboard or mouse directly into a built-in USB port on your laptop instead of an unpowered USB hub, or ensure your external hub is connected to a power supply.
5. **USB Ports Not Working**
   - **Solution:** Connect the device to another USB port and check if it works.
6. **USB Ports Disabled in BIOS/UEFI Settings**
   - **Solution:** Restart your laptop, enter the **BIOS/UEFI setup** (by pressing keys like F2, F12, or Delete during boot), and verify that legacy **USB Keyboard/Mouse Support** is enabled.

2. Simulate a troubleshooting flow: Your Bluetooth headphones are not connecting to your phone. Write the exact step-by-step actions you would take to diagnose and fix the issue, mentioning any settings or indicators you would check.

Here is the exact step-by-step troubleshooting flow to diagnose and fix a Bluetooth headphone connection failure on a mobile phone:

Step 1: Check Physical Indicators & Power

- **Action:** Press and hold the power button on the headphones.
- **Indicators to check:** Look at the LED indicator light.
  - If there is **no light**, the battery is completely dead. Place them in the charging case or plug them into a charger for 10 minutes before retrying.
  - If it flashes a **slow, single color (usually blue or white)**, it is powered on but looking for its last paired device.
  - If it flashes **alternating colors (red/blue) or blinks rapidly**, it is actively in pairing mode.

Step 2: Cycle Phone Bluetooth & Check Airplane Mode

- **Action:** Swipe down/up to open your phone's **Quick Settings / Control Centre**.
- **Settings to check:**
  - Ensure **Airplane Mode** is turned off.
  - Tap the **Bluetooth icon** to turn it off, wait 5 seconds, and tap it again to turn it back on. This resets the phone's local Bluetooth radio broadcast.

Step 3: Evict Conflicting Connections

- **Action:** Look around your immediate area for other devices you frequently use (like your laptop, tablet, or a partner's phone).
- **Reasoning:** Bluetooth headphones usually auto-connect to the *last used device*. If your headphones are secretly connected to your laptop in the next room, they will refuse to connect to your phone. Turn off Bluetooth on those secondary devices temporarily.

Step 4: Force Pairing Mode manually

- **Action:** Do not rely on the headphones automatically connecting. Force them into **Pairing Mode**.
- **How-to:** With the headphones turned on, press and hold the power button (or the dedicated Bluetooth button on the charging case) for 5–7 seconds until the LED light flashes rapidly.

Step 5: "Forget" and Re-pair the Device

- **Action:** Clean out the old connection cache on your phone.
- **Settings to check:**
  1. Go to **Settings > Bluetooth**.
  2. Look under the "Paired Devices" or "My Devices" list.
  3. Tap the **gear icon** or **"i" icon** next to the headphones' name.

4. Select **Forget This Device** or **Unpair**.

5. Look down at the **"Available Devices"** list, wait for the headphones' name to reappear, and tap it to establish a fresh connection handshake.

Step 6: The Ultimate Reset (If still failing)

- **Action:** If the phone says "Pairing unsuccessful," perform a hardware factory reset on the headphones.

- **How-to:** (Varies by brand, but typically involves putting the buds in the case and holding the case button for 15 seconds until the light flashes red). Then repeat Step 5.

3. A user reports that their external webcam is not detected during a Zoom call on Windows 10. Write a checklist of 6 things to check or try to resolve the problem.

Hint: Consider device manager, privacy settings, and app permissions.

Here is a 6 steps trouble shooting to resolve external webcam detection issues on Windows 10:

- **Verify the Physical Connection**: Disconnect the webcam and plug it into a different USB port (preferably a USB 3.0 port directly on the motherboard, not a hub). Ensure the cable is securely seated and the camera lens privacy shutter is fully open.

- **Check Windows Camera Privacy Settings**: Open the Windows Start Menu, go to **Settings > Privacy > Camera**. Ensure that **"Camera access for this device"** is turned on and that the toggle for **"Allow apps to access your camera"** is set to Enabled.

- **Grant Zoom App Permissions**: In the same Camera Privacy menu, scroll down to "Choose which Microsoft Store apps can access your camera" and **"Allow desktop apps to access your camera."**Make sure the toggle next to **Zoom Meetings** is turned on.

- **Inspect Device Manager for Driver Issues**: Right-click the Start button and select **Device Manager**. Expand the **Cameras** or **Imaging devices** section; if you see a yellow warning triangle next to your webcam, right-click it, select **Update driver**, or choose **Uninstall device** and restart your PC to let Windows reinstall it automatically.

- **Select the Correct Camera in Zoom**: Open the Zoom desktop client, click your profile picture, and go to **Settings > Video**. Under the **Camera** dropdown menu, ensure your external webcam is selected instead of an integrated webcam or a virtual camera driver.

- **Close Conflicting Applications**: Check if other software that utilizes the camera (such as Skype, Microsoft Teams, Discord, or the Windows Camera app) is open. **Close these applications completely**via the Task Manager, as Windows typically allows only one application to stream from a webcam at a time.

- **Last Option : Uninstall Zoom**, and **Restart your Computer** And **Reinstall** the latest Version of Zoom from Download Centre. Always Update the Latest version of Zoom App.

4. You are helping a friend whose printer is showing 'offline' status on their PC. Describe how you would use the Windows Troubleshooter tool to try to fix the issue, and explain what information the tool provides during the process.

Step 1 : Open Windows Settings

- Click the Start Menu and select Settings.
- Go to Update & Security.

Step 2 : Open Troubleshooter

- Select Troubleshoot from the left-side menu.
- Click on Additional troubleshooters.
- Select Printer and click Run the troubleshooter.

Step 3: Select the Printer

- Windows will detect the connected printers.
- Select the printer showing an offline status and continue.

Step 4 : Let Windows Diagnose the Problem

- The troubleshooter will check for common problems, such as:

- Printer connection issues.
- Incorrect printer settings.
- Printer driver problems.
- Print Spooler service issues.
- Printer being paused or set to offline mode.

Step 5 : Apply Suggested Fixes

- Windows will display the problems it finds and suggest possible solutions.
- Click Apply this fix if available and follow the instructions.

Step 6 : Test the Printer

- Restart the printer and computer if required.
- Try printing a test page to check whether the printer is back online.

What Information Does Windows Troubleshooter Provide?

The Windows Troubleshooter provides:

1. **Detected Problems:** Identifies possible reasons why the printer is offline.
2. **Diagnostic Results:** Shows which printer settings or services are not working correctly.
3. **Suggested Solutions:** Recommends steps to fix detected problems.
4. **Fix Status:** Tells whether the issue was successfully resolved or still requires manual troubleshooting.

Conclusion: Windows Troubleshooter helps identify and resolve common printer problems automatically, saving time and reducing the need for manual troubleshooting.